

Use Percent to Solve Problems

Lesson 4-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*20% off \$50 → discount is \$10***Discount**

Money taken off the first price to make it cheaper.

*A \$30 item with 10% markup sells for \$33***Markup**

Money added to the cost to set the selling price.

*6% tax on \$20 adds \$1.20***Tax**

Extra money added to a price for the government.

*15% tip on \$40 = \$6***Tip**

Extra money you give for good service.

*50% means 50 out of 100***Percent**

A way to compare a number to 100, shown with the % sign.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can solve problems involving discounts, markups, tax, and tips using percents.

1 Notice

Welcome to the arcade prize shop!

2 Key idea

I can solve problems involving discounts, markups, tax, and tips using percents.

3 Words I'll use

Discount

Markup

Tax

Tip

Percent



Think About It

- What happens to the price when there is a discount?
- What happens to the price when tax is added?
- Does it matter whether you apply the discount or tax first?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A prize costs 120 tickets and is 25% off. What is the sale price?

- A. 90 tickets
- B. 95 tickets
- C. 100 tickets
- D. 30 tickets

Step 1 $25\% \text{ of } 120 = 0.25 \times 120 = 30.$

Step 2 Sale price: $120 - 30 = 90$ tickets.



Answer: A. 90 tickets

 We do — together

Solve this one with your class using the same steps.

Problem: A meal costs \$20. You leave a 15% tip. How much is the tip?

- A. \$3.00
- B. \$1.50
- C. \$5.00
- D. \$0.15

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A shirt costs \$40 and the sales tax is 5%. What is the total cost including tax?

- A. \$42.00
- B. \$45.00
- C. \$40.05
- D. \$38.00

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

At the prize shop a bear is 25% off but the shop adds 5% tax. Which of these raises the price and which lowers it, and what do the words discount, markup, tax, and tip have in common?

Level 1 support

Start here: A discount lowers the price, while tax, tip, and markup raise it.

Need a hint?

Sentence starters (optional)

A ____ makes the price go down, but a ____ makes it go up.

Un ____ hace que el precio baje, pero un ____ hace que suba.

They are alike because each is a ____ of the price.

Se parecen porque cada uno es un ____ del precio.

Word bank: discount markup tax tip percent

Listen for: Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.

Level 2

Does it matter whether the shop takes the discount first or adds the tax first? Predict, and explain your reasoning before you calculate.

- I think the order does / does not matter because ____.
- Applying the discount first means ____.

EXPLORE

In your table you found the discount, then the tax. Why do we calculate the tax on the price AFTER the discount instead of the original price?

Level 1 support

Start here: Tax is charged on the lower price you actually pay after the discount.

 **Need a hint?**

Sentence starters (optional)

I find the discount first because ____.

Primero hallo el descuento porque ____.

Then I take 5% tax on the ____ price, which is ____.

Luego tomo 5% de impuesto sobre el precio ____, que es ____.

Word bank: discount tax percent markup

Listen for: Students explain the discount lowers the price first, then tax is applied to that new lower price (e.g., 5% of 72 = 3.6).

Level 2

A classmate added the 5% tax and the 25% discount into one '20% off' step. Explain why combining them like that gives the wrong final price.

- Combining them is wrong because ____.
- Tax and discount cannot just be added because ____.

CONNECT

An arcade membership is \$50, marked 30% off, then 6% tax is added. Your friend says the total is \$37.10. Is that right, and how would you prove it?

Level 1 support

Start here: After 30% off the price is \$35, then 6% tax makes the total \$37.10.

 Need a hint?

Sentence starters (optional)

First the discount: 30% of \$50 = \$____, so the sale price is \$____.

Primero el descuento: 30% de \$50 = \$____, así que el precio rebajado es \$____.

Then 6% tax on \$____ adds \$____, for a total of \$____.

Luego 6% de impuesto sobre \$____ suma \$____, para un total de \$____.

Word bank: discount tax percent tip

Listen for: Students compute \$15 discount, \$35 sale price, \$2.10 tax, and \$37.10 total, confirming the friend is correct with shown steps.

Level 2

Suppose the membership were a tip situation instead: \$50 plus a 20% tip. Compare that total to the \$37.10 discount-then-tax total and explain why one is so much higher.

- A 20% tip makes the total \$____ because ____.
- It is higher than \$37.10 because ____.

Write About the Math

The Writing Revolution

I can explain my solution using the words discount, markup, tax, tip, and percent.

1. Kernel Sentence subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.

Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.

Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Percent matters in math

Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.

Porcentaje importa en matemáticas porque ____.

but
pero

Percent matters in math, but ____.

Porcentaje importa en matemáticas, pero ____.

so
entonces

Percent matters in math, so ____.

Porcentaje importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about percent.

Di un hecho verdadero sobre percent.

Percent ____.

Question *Pregunta*

Ask a question about percent.

Haz una pregunta sobre percent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about percent.

Muestra entusiasmo sobre percent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with percent.

Dile a un compañero qué hacer con percent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

A ____ makes the price go down, but a ____ makes it go up.

Un ____ hace que el precio baje, pero un ____ hace que suba.

They are alike because each is a ____ of the price.

Se parecen porque cada uno es un ____ del precio.

I find the discount first because ____.

Primero hallo el descuento porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A test has 25 questions. A student gets 88% correct. How many are correct?

- A. 22
- B. 20
- C. 23
- D. 21

Show your work:

2. A shirt is \$30 after a 25% discount off the original price. What was the original price?

- A. \$40
- B. \$37.50
- C. \$45
- D. \$36

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A store has a 20% off sale, then takes an additional 10% off the sale price. Is this the same as 30% off the original? Use \$100 as the original price. Show your work for both scenarios and explain the difference.

Sentence starter: Taking 20% off then 10% off gives a final price of \$____, but 30% off the original gives \$____. The difference is \$____ because ____.

Show your work:

Reflect — Exit Ticket

A jacket costs \$80 and is 35% off. Sales tax is 5%. What is the final price?

- A. \$54.60
- B. \$52.00
- C. \$56.00
- D. \$49.40

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $\$3.00 - 15\% \text{ of } \$20 = 0.15 \times 20 = \$3.00$.
2. **You Try (notes):** A. $\$42.00 - \text{Tax: } 0.05 \times \$40 = \$2.00$. Total: $\$40 + \$2 = \$42.00$.
3. **Try It 1:** A. $22 - 0.88 \times 25 = 22 \text{ questions}$.
4. **Try It 2:** A. $\$40 - \$30 \text{ is } 75\% \text{ of the original; } 30 \div 0.75 = \40 .
5. **Exit Ticket:** A. $\$54.60 - \text{Discount: } 0.35 \times \$80 = \$28$. Sale price: $\$80 - \$28 = \$52$. Tax: $0.05 \times \$52 = \2.60 . Final: $\$52 + \$2.60 = \$54.60$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).